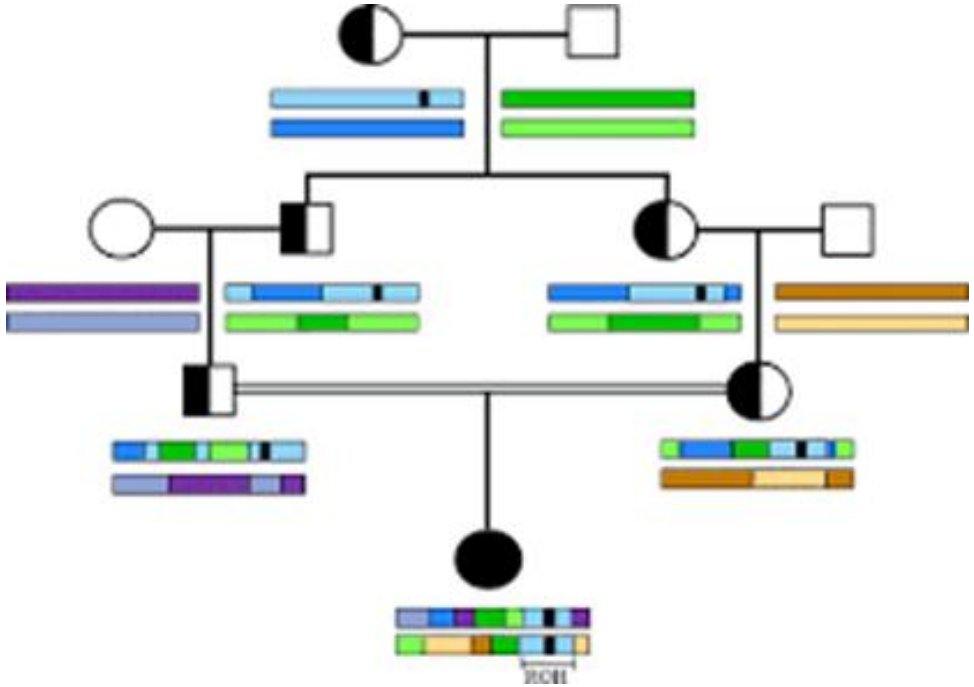
Runs of Homozygosity (ROH) in Humans genome research

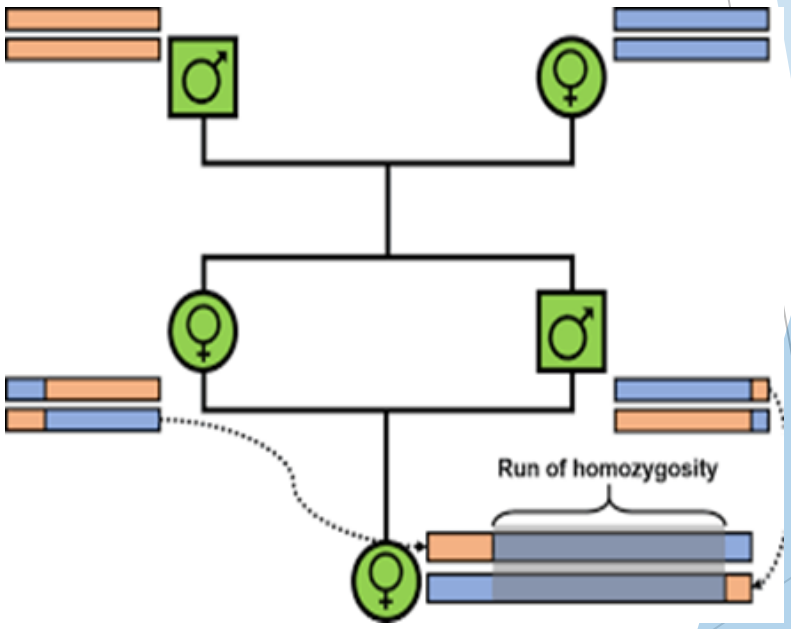
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What is ROH?

Defined as contiguous homozygous segments arising from shared ancestry, provide valuable insights into the demographic history, inbreeding levels, and selection pressures within populations.





Terms:

اصطلاح	توضیح
ROH	ناحیه هموزیگوت در ژنوم
FROH	درصد ژنوم داخل ROH
Long ROH	نشانه recent inbreeding
Short RO	نشانه ancient ancestry
ROH hotspot / island	ناحیه پرتکرار ROH
Inbreeding depression	کاهش fitness به علت inbreeding

FROH: Fraction of the genome in ROH

- **Measuring Inbreeding and Demography with ROH**
- **FROH:** Fraction of the genome in ROH is a robust genomic inbreeding measure, correlating strongly with pedigree inbreeding (r ≈ 0.86) in Europeans (McQuillan et al. 2008, 359-72 ; Shafer and Kardos 2025; Curik et al. 2014, 26-34).
- ROH profiles classify populations by recent consanguinity vs. ancient isolation (Pemberton et al. 2012, 275-92 ; Kirin et al. 2010; Nalls et al. 2009; Minn et al. 2025, 287 - 296).
- In European-ancestry North Americans, autozygosity and ROH burden have declined over the 20th century, consistent with increasing outbreeding and urbanization (Nalls et al. 2009).

فرمول کلی:

$$F_{ROH} = \frac{\sum ROH\ length}{Total\ autosomal\ genome\ length}$$

- FHOM: excess of homozygosity-based inbreeding coefficient

$$\left(F = \frac{\text{Observed Hom} - \text{Expected Hom}}{1 - \text{Expected Hom}}\right)$$

ROH in human

Examples of Population Patterns

Population context	ROH pattern (simplified)
South/West Asians with frequent cousin marriage	Many very long ROH
Oceanians (isolated, small Ne)	Many short ROH, few long
Native Americans	Many short and long ROH
Urban Europeans, East Asians	Mostly background short ROH

ROH and Disease / Trait Architecture

- Long ROH are **enriched for deleterious homozygous variants**, especially damaging coding alleles (Szpiech et al. 2013, 90-102 ; Szpiech et al. 2018, 747 - 762; Pan and Szpiech 2025).
- This likely explains associations between higher ROH burden and increased risk for complex diseases and inbreeding depression traits (e.g., schizophrenia, Alzheimer's, cancers, height, cognition) (Ceballos et al. 2018, 220-234; Szpiech et al. 2013, 90-102 ; Ceballos et al. 2017; Szpiech et al. 2018, 747 - 762; Pan and Szpiech 2025).
- In admixed groups (African American, Puerto Rican, Mexican American), deleterious homozygotes occur more often when ROH overlap African-ancestry segments, suggesting ancestry-specific impacts on complex disease risk (Szpiech et al. 2018, 747 - 762).

Methods and New Directions

- ROH are detected from SNP arrays and whole-genome sequencing; thresholds, SNP density, and software (e.g., PLINK, model-based methods) strongly influence results (McQuillan et al. 2008, 359-72 ; Ceballos et al. 2017; Shafer and Kardos 2025; Lavanchy and Goudet 2022; Minn et al. 2025, 287 - 296).
- New tools (e.g., ROH-DICE, ROH-based machine learning, clinical ROH pipelines) allow locus-specific disease mapping and consanguinity prediction in large cohorts and clinical WES data (Zeitler and Gilbert 2024; Naseri et al. 2024; Valente et al. 2024).

Conclusion

- In humans, runs of homozygosity are widespread genomic footprints of shared ancestry.
- Their number and length reveal recent inbreeding, older demographic events, and cultural patterns, while their content helps explain recessive disease risk and complex trait variation. ROH-based inbreeding measures now complement or surpass pedigrees, making ROH a central tool in human population genetics and medical genomics.

Thank you for your attention